

Matrix Factorization

$$Ax = b$$

$$L(Ux) = b$$

$$Ux = y$$

$$Ly = b$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix} \begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

L U

A is $m \times n$

L is $m \times m$
lower triangular matrix with 1's on the diagonal

U is $m \times n$
echelon form of A



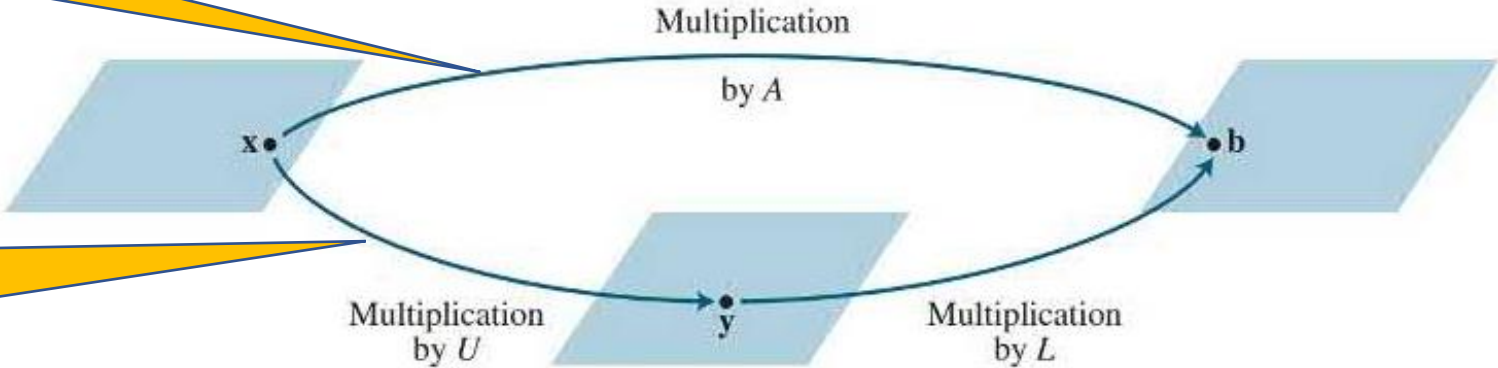
$$\begin{matrix} Ly = b \\ Ux = y \end{matrix}$$

(2)

The ordinary way (one step)

First solve $Ly = b$ for y , and then solve $Ux = y$ for x . See Figure 2. Each equation is easy to solve because L and U are triangular.

The Matrix Factorization (two steps)



Example:

$$A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} = LU$$

Use this LU factorization of A to solve $A\mathbf{x} = \mathbf{b}$, where $\mathbf{b} = \begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix}$.

Step 1: Find $\mathbf{y}$

$$L\mathbf{y} = \mathbf{b} \quad \begin{bmatrix} L & \mathbf{b} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} -9 \\ -4 \\ 5 \\ 1 \end{bmatrix}$$

Augmented matrix

The Forward algorithm to get echelon form:

Submatrix 1: $R_2 = R_2 + R_1 \rightarrow R_3 = R_3 - 2R_1 \rightarrow R_4 = R_4 + 3R_1$

Submatrix 2: $R_3 = R_3 + 5R_2 \rightarrow R_4 = R_4 - 8R_2$

Submatrix 3: $R_4 = R_4 - 3R_3$

Step 2: Find $\mathbf{x}$

$$U\mathbf{x} = \mathbf{y} \quad \begin{bmatrix} U & \mathbf{y} \end{bmatrix} = \begin{bmatrix} 3 & -7 & -2 & 2 & -9 \\ 0 & -2 & -1 & 2 & -4 \\ 0 & 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 3 \\ 4 \\ -6 \\ -1 \end{bmatrix}$$

Augmented matrix

The Backward algorithm to get reduced echelon form:

C4: $R_4 = -R_4 \rightarrow R_3 = R_3 - R_4 \rightarrow R_2 = R_2 - 2R_4 \rightarrow R_1 = R_1 - 2R_4 \rightarrow \underline{C4} \checkmark$

C3: $R_3 = -R_3 \rightarrow R_2 = R_2 + R_3 \rightarrow R_1 = R_1 + 2R_3 \rightarrow \underline{C3} \checkmark$

C2: $R_2 = R_2 / -2 \rightarrow R_1 = R_1 + 7R_2 \rightarrow \underline{C2} \checkmark$

C1: $R_1 = R_1 / 3 \rightarrow \underline{C1} \checkmark$

Total arithmetic operations in both Step 1 and 2 is around **28 operations**. (not counting LU factorization)

If we go the ordinary way (just in one step), it takes **62 operations** !!

But how do we get U and L ? See the next slide

An LU Factorization Algorithm

EXAMPLE 2 Find an LU factorization of

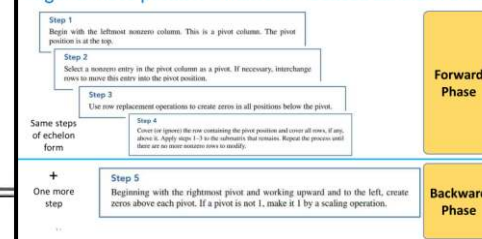
$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

ALGORITHM FOR AN LU FACTORIZATION

1. Reduce A to an echelon form U by a sequence of row replacement operations, if possible.
2. Place entries in L such that the *same sequence of row operations* reduces L to I .

Recall this algorithm from Chapter 1

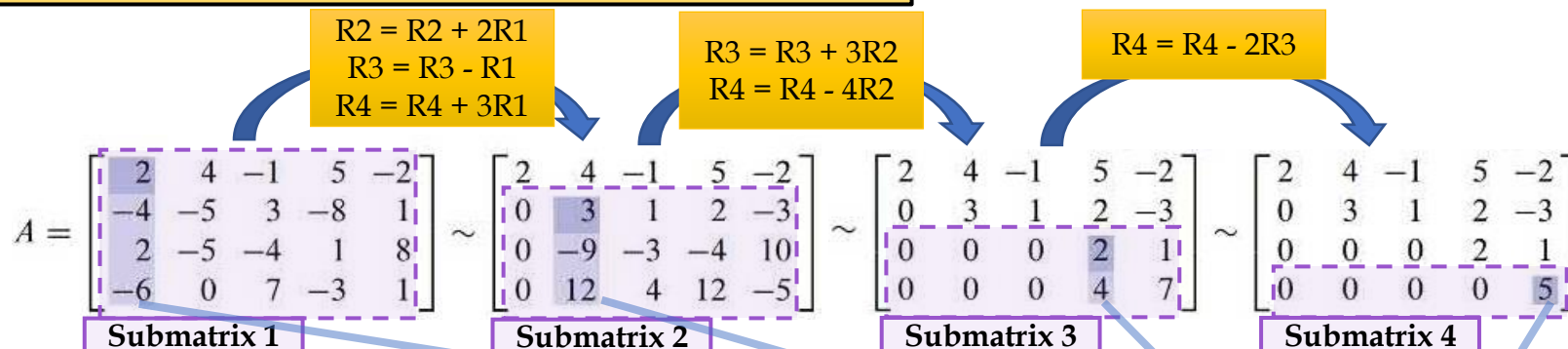
Algorithms to produce a matrix in Reduced Echelon Form



Important:

To check your work multiply LU (NOT UL) and get A

Step 1: Apply The Forward algorithm to get echelon form:



Step 2: At each pivot column in each submatrix, divide the highlighted entries by the pivot and place the result into L :

Step 3: zero padding

Important:

In some cases, you may get less pivot columns than you need to make L . Just put zeros at the missing columns, but the diagonal you MUST put 1

Remember:

L is always a square matrix with size equal to the number of rows in A . in this example L is 4x4.

$$\begin{bmatrix} 2 \\ -4 \\ 2 \\ -6 \end{bmatrix} \begin{bmatrix} 3 \\ -9 \\ 12 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \begin{bmatrix} 5 \end{bmatrix} \begin{matrix} \div 2 \\ \div 3 \\ \div 2 \\ \div 5 \end{matrix} \begin{matrix} \downarrow \\ \downarrow \\ \downarrow \\ \downarrow \end{matrix} \begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ 1 & -3 & 1 & \\ -3 & 4 & 2 & 1 \end{bmatrix}, \text{ and } L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$